

LONNIE ERNST

Biotechnologist, Molecular Biology, & Biomanufacturing

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PROFESSIONAL SUMMARY

Biotechnologist with a B.S. in Biotechnology and laboratory experience spanning academic and GMP manufacturing settings across molecular biology, nucleic acid extraction and quantification, PCR, and microbiology. Background combines hands-on research at an academic biomanufacturing lab with reagent formulation in a GMP production environment. Driven by the positive impact biotechnology can have on environmental and human health, with a growing focus on bringing data analysis and computational tools into laboratory work.

PROFESSIONAL EXPERIENCE

Specialist III, Media Production Coordinator | Salt Lake Community College InnovaBio | Sep 2021 - Present

- Perform high-complexity molecular biology procedures including PCR, nucleic acid extraction and quantification, agarose gel electrophoresis, restriction digest, NanoDrop quantification, and pH measurement.
- Participate in development, optimization, and validation of new assays and protocols; write and maintain SOPs and troubleshoot assays.
- Train laboratory personnel in molecular techniques, contamination control, equipment operation, data recording, and results analysis.
- Lead media production for biology departments, producing 1,000+ orders per semester across 15+ media types with full QC including viability and sterility controls.
- Manage inventory, vendor communications, equipment purchases, and cost-tracking systems including production cost analysis and billing.

Formulator II | bioMerieux | May 2023 - Present

- Perform aseptic formulation of buffers, reagents, and PCR master mix components for molecular diagnostic platforms in a high-throughput GMP manufacturing environment, processing up to 36,000 pouches per day.
- Formulate and dispense magnetic beads using a BioTek MultiFlo automated dispenser; operate and troubleshoot freeze dryers.
- Complete batch records, verify protocol compliance, manage bi-weekly inventory, and maintain contamination control through rigorous GMP practices.
- Train Form 1 personnel in GMP documentation, reagent preparation, and contamination-control standards.

Microbiologist | Analytical Research Labs | May 2022 - May 2023

- Performed high-throughput aseptic sample preparation and microbial enumeration for nutraceutical, food, and supplement manufacturers, processing 100+ samples per day.
- Prepared positive plate controls using live pathogens; maintained BSL-2 aseptic and sterile technique throughout.
- Served as night shift team lead, directing sample preparation team to meet testing deadlines under time pressure.
- Tracked and documented samples through full chain of custody using LIMS software.

EDUCATION

Bachelor of Science, Biotechnology | Utah Valley University | May 2026

TECHNICAL SKILLS

- **Molecular Biology:** PCR, Primer Design, Nucleic Acid Extraction & Quantification, Gel Electrophoresis, Restriction Digest, DNA Purification, Plasmid Design, Serial Dilution, Aseptic & Sterile Technique, Microbial Culture & Enumeration, Protein Expression & Purification, BSL-2 Operations
- **Biomanufacturing & Production:** GMP Manufacturing, Reagent & Buffer Formulation, PCR Master Mix Formulation, Automated Dispensing Equipment, Media Production (15+ types), Freeze Drying, High-Throughput Pipetting, Batch Record Completion
- **Instruments & Equipment:** NanoDrop, DeNovix DS-11 FX+, BioTek MultiFlo, MediaClave, Integra MediaJet Plate Pourer, Freeze Dryers, Photobioreactor, Anaerobic Chamber, pH Probe, Autoclave, Thermocycler, UV Transilluminator
- **Data & Quality:** SOP Writing & Execution, Assay Development & Validation, Quality Control, LIMS, GMP Compliance, Data Analysis & Visualization (R), Command Line & Bash Scripting, Electronic Lab Notebook